Harshvivek Kashid

Indian Institute of Technology Bombay (India)
+91-9082055427 • harshvivek14@gmail.com
scholar.google.com/citations?user=KzyJHVYAAAAJ
github.com/harshvivek14 linkedin.com/in/harshvivek-kashid

Program	Institution	CGPA	Year
M.Tech. (Computer Science & Engg.)	Indian Institute of Technology Bombay	**9.33/10**	2023-25
B.E. (Information Technology)	Ramrao Adik Institute of Technology	**9.39/10**	2018-22

Work Experience

1. *Machine Learning Researcher, Deccan AI, Hyderabad* **May 2026 - Present**

- Working on high-impact problems in the areas of **Reinforcement Learning (RL)** and **evaluation loops** of LLMs and agentic systems (coding and non-coding domains); building and certifying **RL environments with verifiable rewards**, and analyzing large-scale frontier-model rollouts to locate real capability gaps.

2. *Data & AI Engineer, Nexus Cognitive, Pune* **July 2025 - May 2026**

- Designed and deployed **AIOps platform** using **LLMs, RAG, and LangGraph**, with scalable AI services implemented in **FastAPI** that automated alert triage and analysis, saving ~**90% of engineering effort**.
- Integrated observability streams via Datadog **MCP** (logs, metrics, alerts) and Notion MCP enabling context enrichment, semantic summarization, and automated root cause analysis.
- Implemented **vector-based retrieval pipelines** using **FAISS** to ground LLM reasoning in historical incidents and operational knowledge, reducing hallucinations and improving alert relevance.

3. *Software Development Engineer, Amazon, Bengaluru* **July 2022 - July 2023**

- Co-led the end-to-end implementation of the Digital Software & Video Games category, collaborating with cross-functional team. Owned the system design and development of **1-Click Digital Checkout** to support Buy-Now functionality for digital orders in India and seamless onboarding of 7+ global marketplaces with **95%** code reusability.
- Worked on Low-Level Design for **Grid Layout** for books on Amazon Search Page in 10+ global marketplaces.
- Engineered the **R4A** (Reconciliation for Amazon) Accounting tool onboarding to 10+ marketplaces with four stages of transaction processing: Extraction (ETL), Unification, Classification and Dashboarding.
- Received **four accolade awards** for design documents, code reviews, bug resolutions, and teamwork.

4. *Research Intern, Hyundai Motors Engineering, Hyderabad* **June 2024 - July 2024**

- Conducted research to enhance Car-AI by designing follow-up question generation and knowledge-grounded QA pipelines, with automatic evaluation using LLM-as-a-judge, work published at **ACL 2025**.

5. *SDE Intern, ElasticRun, Pune* **Feb 2022 – June 2022**

- Contributed to backend engineering for operations and user-profile dashboards by designing and implementing **RESTful APIs** in **Python** using the **Frappe Framework** and **Postgres** relational database.

Research Publications

1. *Follow-Up Questions Generation Using Bloom's Taxonomy and Evaluation using Grice's Maxims*

- Research paper at **ACL 2025** on improving multi-turn conversational reasoning in Car AI assistants.
- Proposed **B-FQG**, a cognitively grounded follow-up question generation framework that operationalizes **Bloom's Taxonomy** to systematically escalate question complexity from recall to creation.
- Developed **GriceWise**, a **reference-free evaluation framework** based on **Grice's Maxims** (Logical Consistency, Informativeness, Relevance, Clarity) to assess follow-up question quality beyond surface fluency.
- Curated and released **Blooms-FQ**, a large-scale human-annotated dataset with 750 seed questions and 3,750 follow-ups; achieved **96.68% human agreement** with intended cognitive levels.
- Identified the cognitive complexity thresholds at which in-car conversational agents begin to degrade, providing actionable diagnostics for optimizing multi-turn LLM-based assistants.

2. *RoundTripOCR: Post-OCR Error Correction using Machine Translation Approach*

- Published at the Proceedings of the 21st **International Conference on Natural Language Processing**, 2024.

- Proposed **RoundTripOCR**, a novel approach to artificially generate Post-OCR error correction data for Devanagari languages in the form $<T', T>$, where T' is the OCR output and T is the corrected text. (**25 million** datapoints).
- Established new benchmarks for the Post-OCR error correction task for **six low-resource** Devanagari languages based on the pre-trained Seq2Seq language models: **mBART, mT5** and **IndicBART**.

3. *Legal Large Language Models with NLSUI and Meta AI: Grahak-Nyay & Nyay-Darpan*

- Research work on domain-adapted Large Language Models (LLMs) for Indian consumer law, in collaboration with NLSUI and Meta AI; **both papers accepted at AACL 2025 (JUSTNLP Workshop)**.
- **Grahak-Nyay**: Designed and implemented an end-to-end Retrieval-Augmented Generation (RAG) system for consumer grievance redressal using open-source LLMs, LangChain orchestration, and hybrid dense-sparse legal document retrieval.
- Built scalable document ingestion, chunking, and embedding pipelines over statutes, consumer court judgments, and regulatory guidelines to improve factual grounding, citation accuracy, and hallucination mitigation.
- **Nyay-Darpan**: Developed a two-stage LLM framework combining long-document summarization and semantic case-law retrieval to support AI-assisted decision-making in consumer dispute resolution.
- Applied prompt tuning, Chain-of-Thought (CoT) prompting, and task-specific reasoning templates to improve legal reasoning accuracy, interpretability, and consistency in generated summaries, along with **LLM-as-a-Judge** evaluation.

4. *HalluDetect: Detecting, Mitigating, and Benchmarking Hallucinations in Conversational Systems*

- Research paper at **EMNLP 2025** on hallucination detection and mitigation in LLM-based systems (legal domain).
- Developed **HalluDetect**, an LLM-based hallucination detection framework for consumer grievance chatbots built on **LLaMA 3.1 8B Instruct**, achieving **68.92% F1** and outperforming baseline detectors by **22.47%**.
- Benchmarked five hallucination mitigation architectures and identified **AgentBot** as the most effective strategy, reducing hallucinations to **0.4159 per turn** while preserving **96.13% token-level accuracy**.
- Demonstrated that optimized inference-time strategies and agentic reasoning significantly improve factual grounding and reliability of LLMs in high-stakes legal applications.

Technical Skills

- Programming Languages: C, C++, Java, Python, JavaScript, Shell Script
- Frameworks: Flask, PyTorch, FastAPI, TensorFlow, NodeJS, MySQL, Postgres, Terraform, Apache Airflow, Spark, Kafka, LangChain, LangGraph, CrewAI, MCP
- Tools and DevOps: AWS, Azure, Kubernetes, Docker, Datadog, Grafana, Prometheus, MLOps

Positions of Responsibility

- *Teaching Assistant* for CS217 Artificial Intelligence and Machine Learning (Prof. Pushpak Bhattacharyya), IIT Bombay (Jan 2025 – May 2025).
- *Teaching Assistant* for CS626 Speech, Natural Language Processing and the Web (Prof. Pushpak Bhattacharyya), IIT Bombay (Aug 2024 – Dec 2024).
- *Student Companion* at Institute Student Companion Program (ISCP), IIT Bombay (July 2024 – Present).

Achievements/Awards

- Awarded the **Masters Thesis Excellence Award (2025)** for the thesis titled *Large Language Models for Conversation and Evaluation*.
- Secured **All India Rank 57** in *GATE 2022 (CS)* among 77,257 candidates.
- Secured **2nd position** in a 48-hour Hackathon conducted by *RAIT ACM Student Chapter*.
- Runners-up in *Codethon 2022* conducted by CodeChef-RAIT & ACM VIT-AP.
- Ranked in **Top 100** in the National Student Coding Contest conducted by *ACM India Council*.
- Successfully completed the *Amazon ML Summer School Program* in 2024.
- Microsoft Certified: **Azure Data Scientist Associate** certification
- **BCPIS**: Blockchain-based Counterfeit Product Identification System in *Journal of Applied Security Research* (2023).